DeepScientist

AI REVIEW REPORT

AIG_CUR_2026_FARS_chunked_budget_allocation

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Final Agent Report

Structured markdown synthesized by the annotation agent.

Summary

Final Review Report

Summary

Paper type: short empirical/methods note on *world-model verification* for web agents facing irreversible actions.

What the paper claims: Sequential verify-and-retry verification can exhibit **non-monotonic regressions** as verification budget increases, because verifier rejections trigger re-browsing and "trajectory drift" toward worse products (Page 1 - Abstract/Introduction; Page 5 - Failure Analysis). The authors propose **chunked budget allocation**: spend a fixed budget `B` in fewer verification cycles `M` with more parallel rollouts per cycle `K=B/M` (Page 1 - Introduction; Page 3 - Verification Framework). On WebShop, they report a large gain at the same budget (`B=10`): **21.50%** success for chunked no-aggregation (`M=1,K=10`) vs **0.86%** for sequential verify-and-retry (`M≈10,K=1`) (Page 4 - Main Results). They further argue **consensus aggregation fails** because the verifier is severely under-confident (`hat p=0` in 98.4% of cycles) (Page 6 - Calibration Analysis).

My assessment: The core observation—*sequential verifier rejections can harm long-horizon agent trajectories for irreversible commits*—is practically important and plausibly general. However, the current manuscript does not yet support several of its strongest causal conclusions because key parts of the evaluation are **confounded/underspecified** (Page 3 - Experimental Conditions; Page 4 - Main Results): the decision rules and thresholds differ across conditions, the "budget exhausted → auto-pass" rule is strong and not ablated, and there is no **no-verification baseline** (`B=0`) to contextualize how much verification helps beyond "stop rejecting repeatedly." Mechanism analyses (trajectory drift, calibration) are promising but need more rigorous definitions and unbiased reporting (Page 5 - Failure Analysis; Page 6 - Calibration Analysis).

Bottom line: With a tighter algorithm definition + fairer controlled comparisons (structure vs decision rule vs threshold) + an expanded experiment/analysis package, this could become a solid "design principle" paper for agentic test-time compute allocation in irreversible-action settings. In its current form, confidence in the headline causal claims is **moderate** despite the large reported effect size.

Strengths

1. **Practical and timely problem framing (irreversible actions + verifier gating).** The paper focuses on a real failure mode in agent deployment: once an irreversible action is taken (checkout), mistakes cannot be undone. Studying verification *as a decision policy* rather than only improving model quality is a useful framing (Page 1 - Abstract/Introduction).
2. **Simple, easy-to-implement intervention with large observed effect.** Re-allocating a fixed query budget from many sequential cycles to fewer cycles with more parallel rollouts is a minimal change that practitioners can implement without retraining models, and the reported effect size is substantial in this setting (Page 3 - Experimental Conditions; Page 4 -

Main Results).

3. **Mechanism-oriented analysis (not just a leaderboard table).** The paper attempts to explain *why* gains happen (trajectory drift after rejection) rather than only reporting success rates (Page 5 - Failure Analysis). This is the right direction for a short "design principle" paper.
4. **Good instinct to diagnose verifier confidence/calibration.** Reporting that `hat p` collapses to 0 in most cycles and connecting this to consensus failure is a valuable diagnostic angle, and it links the work to a broader literature on confidence estimation for LLM judges/verifiers [6][7] (Page 6 - Calibration Analysis).
5. **Clear writing and clean conceptual variables (`B`, `M`, `K`, τ).** The core knobs are understandable and make the paper easy to discuss and extend (Page 3 - Verification Framework).

Weaknesses

1. **Core comparisons are confounded by decision-rule and threshold choices (validity risk).** The paper attributes gains mainly to "cycle reduction," but key conditions differ in *more than* `M`: Condition B uses consensus with a chosen τ, Condition C uses a single-sample rule, and later analyses discuss a different τ regime. This makes it hard to cleanly separate **structure (M)** from **decision rule / calibration brittleness** (Page 3 - Experimental Conditions; Page 4 - Main Results; Page 4 - Cycle Reduction).
2. **Algorithm/protocol is underspecified at decision-critical edge cases.** The "budget exhausted → auto-pass" policy and the exact definition of a "verification cycle" (when consumed; how multiple checkout attempts are handled) are not fully specified, yet can materially change success rates and the existence of the reported non-monotonicity (Page 3 - Verification Framework).
3. **Missing baselines and missing per-condition behavioral stats weaken interpretability.** There is no explicit **no-verification baseline** (`B=0`), and key intermediate statistics (checkout proposal count, block/pass rate, estimated false-negative rate) are not reported. This makes it difficult to tell whether chunking improves *verification quality* or mainly reduces harmful re-planning events (Page 2 - Problem Setup; Page 4 - Main Results).
4. Failure analysis is selection-biased and "trajectory drift" lacks an operational definition. Conditioning on "B succeeds, A fails" can inflate the apparent dominance of drift among those episodes, and the analysis is not repeated for the best-performing chunked variant (C). The taxonomy needs a precise labeling rubric and symmetric pairwise/unconditional reporting (Page 5 - Failure Analysis).
5. **Calibration diagnosis is promising but currently incomplete.** ECE/Brier computation details (cycle vs episode, binning scheme) and a reliability diagram are missing, which is especially important given the highly concentrated `hat p` distribution. The conclusion "benefit comes entirely from structure" is therefore too strong (Page 6 - Calibration Analysis).
6. **Reproducibility gaps for both actor and verifier.** The paper omits key implementation details (actor prompting/context window; verifier fine-tuning data/objective; history truncation), which are necessary to reproduce results and to assess whether miscalibration/false negatives are expected properties of the chosen verifier (Page 3 - Setup).
7. **Limited generality evidence and novelty positioning still feels incremental.** Results are shown only for one environment and one verifier; related work positioning should more explicitly clarify the overlap with prior world-model planning for irreversible actions [1][2] and test-time compute allocation for agents [4][5], and state the residual novelty as a *verification-budget-allocation pathology* and its boundary conditions (Page 2 - Related Work; Page 6 - Conclusion).

Key Issues

Ranked Error Board (Top-5 core defects)

Rank	Core defect	Severity	Research-value impact	Validity risk	Fixability	Confidence	Anchor
1	**Structure vs decision-rule vs threshold confounding**: attribution of gains to cycle reduction is not clean given differing τ/aggregation and the "auto-pass after budget exhaustion" rule	major	high	high	medium	high	Page 3 - Verification Framework/Experimental Conditions; Page 4 - Main Results/Cycle Reduction
2	**Missing baselines + missing behavioral diagnostics**: no `B=0` baseline; no checkout proposal counts/block rates/false-negative estimates	major	high	medium-high	high	high	Page 2 - Problem Setup; Page 4 - Main Results
3	Selection-biased failure analysis + undefined 'trajectory drift': conditioning on (B succeeds, A fails) and qualitative labeling	major	medium-high	medium	medium	medium-high	Page 5 - Failure Analysis

Rank	Core defect	Severity	Research-value impact	Validity risk	Fixability	Confidence	Anchor
4	**Calibration analysis under-specified**: unclear unit of analysis for Brier/ECE; no binning scheme/reliability diagram; over-strong “entirely structural” claim	major	medium	medium	high	medium-high	Page 6 - Why Consensus Aggregation Fails
5	**Reproducibility gaps for actor/verifier**: prompts, history truncation, verifier fine-tuning data/objective not described	major	medium	medium	medium	medium	Page 3 - Setup

Evidence-sufficiency + objectivity audit (what is currently proven vs not)

- **[Proven / well-supported in this setting]** A large performance gap exists between a sequential verify-and-retry configuration and a chunked (`M=1`) configuration at the same `B=10` (Page 4 - Main Results).
- **[Partially proven]** The claim that *cycle reduction* is the dominant causal factor is plausible but not isolated from decision-rule/threshold effects (Page 3 - Experimental Conditions; Page 4 - Cycle Reduction).
- **[Partially proven]** “Trajectory drift” is a plausible mechanism but needs an operational definition and unbiased statistics before being called “dominant” (Page 5 - Failure Analysis).
- **[Partially proven]** Consensus failure is consistent with severe under-confidence, but calibration reporting needs stronger methodology detail (Page 6 - Calibration Analysis).

ASCII Diagram — Paper Structure & Evidence Map

```
[Goal: safer irreversible actions via world-model verification]
   |
   +--> Claim C1: Sequential verify-and-retry can regress as budget increases
   |       Evidence: A vs (chunked) in Table 1; M-sweep trend (Fig 2)
   |       Gap: no B=0 baseline; auto-pass policy + tau choices may alter
monotonicity
   |
```

```
   +--> Claim C2: Chunking budget into fewer cycles (small M) mitigates regression
   |       Evidence: best results at M=1; failure patterns suggest drift after
rejection
   |       Gap: M changes K; consensus rule strictness changes with K and tau
   |
   +--> Claim C3: Consensus aggregation fails due to verifier under-confidence
           Evidence: p-hat=0 for 98.4% of cycles; ECE/Brier reported
           Gap: clarify calibration computation unit + add reliability diagram;
test
alt aggregators

[Main decision risk]
   -> Current experiments cannot fully separate (structure M) vs (decision rule +
tau) vs (budget-exhaustion policy)
   -> Revision should add baselines + factorial controls to make claims
reviewer-proof
```

Actionable Suggestions

Must (publication-critical)

1. **Add an explicit algorithm box for the verification policy family.**
   - Define precisely: what constitutes a cycle; when cycles are consumed; what happens on reject; what happens when budget is exhausted (auto-pass vs abstain vs terminate); and how multiple checkout attempts are handled (Page 3 - Verification Framework).
   - *Acceptance criteria:* a reader can implement A/B/C exactly from the paper without guessing.
2. Unconfound "structure (M)" from "decision rule/threshold."
   - Add a controlled **factorial** experiment: vary `M` while holding the decision rule fixed; vary decision rule while holding `M` fixed; and for consensus, sweep/tune τ for each `K` (Page 3 - Experimental Conditions; Page 4 - Cycle Reduction).
   - Report CIs for each cell.
   - *Acceptance criteria:* you can state "holding decision rule fixed, increasing M decreases success" (or revise claim accordingly).
3. **Add a no-verification baseline and behavior-level diagnostics.**
   - Add `B=0` (no verification) to contextualize whether verification helps at all relative to "just checkout" (Page 2 - Problem Setup; Page 4 - Main Results).
   - Report per-condition diagnostics: number of checkout proposals; number blocked; pass rate; mean steps-to-checkout; estimated false-negative rate (conditional on eventual successful checkout) if definable.
   - *Acceptance criteria:* the paper can decompose gain into "less rejection-induced re-planning" vs "better acceptance decisions."
4. **Make trajectory-drift analysis unbiased and operational.**
   - Provide an explicit rubric for "trajectory drift," "wrong product," "never reached checkout."
   - Report unconditional frequencies + symmetric pairwise conditional sets (A succeeds/B fails, etc.), and run the analysis for **C vs A** since C is best (Page 5 - Failure Analysis).
   - *Acceptance criteria:* "trajectory drift is dominant" is supported by unbiased statistics, or softened.
5. **Strengthen calibration reporting and align claims.**

- Clarify whether calibration is computed over cycles or episodes; specify ECE binning; add a reliability diagram (or cumulative calibration plot) (Page 6 - Calibration Analysis).
- Consider adding a simple calibration/uncertainty baseline (e.g., verbalized confidence) or at least discuss its feasibility, given known calibration pitfalls for LLM judges [6][7].

Nice-to-have (high leverage, but not strictly required)

6. Test a more general "permissive aggregator" family. Under under-confidence, compare consensus to `any-positive`, `max`, or "pass if at least one rollout predicts success" as a more appropriate decision rule class than majority voting.
7. **Generalization tests.** Add (i) a second domain with irreversible actions (e.g., a WebArena subset), and/or (ii) a second verifier with different calibration profile (Page 6 - Conclusion). If too costly, run a smaller-scale study (e.g., 50–100 tasks) to check directionality.
8. **Writing defensibility sweep.** Replace absolute causal language ("not accuracy," "entirely structural," "dominant") with evidence-consistent scoped claims where isolation is incomplete (Page 1 - Introduction; Page 6 - Calibration Analysis/Conclusion).

ASCII Diagram — Revision Strategy Roadmap

```
[Confounding: M vs decision rule vs tau vs auto-pass]
  -> Add Algorithm 1 + explicitly define policies
  -> Add factorial experiment (M × decision rule × tau)
  -> Expected gain: causal attribution becomes reviewer-proof

[Missing baselines + missing diagnostics]
  -> Add B=0 baseline + per-condition checkout/pass/block stats
  -> Expected gain: interpretability; shows whether verification truly helps

[Mechanism claims not yet unbiased]
  -> Define drift rubric + symmetric/unconditional failure analysis
  -> Expected gain: mechanism becomes credible and reusable

[Calibration story incomplete]
  -> Add reliability diagram + clarify unit + consider calibration baseline
  -> Expected gain: consensus-failure claim becomes technically defensible
```

Storyline Options + Writing Outlines

Storyline options (pick one and rewrite Title/Abstract/Intro accordingly)

Option A (recommended): "Compute allocation for verifier-gated irreversible commits."

- Arc: Irreversible commits are safety-critical → verifier-gating seems natural → sequential verify-and-retry can *harm* as budget increases → root cause is rejection-induced trajectory drift → key fix is allocate compute into fewer sequential interventions → aggregation only helps when verifier is calibrated.
- Best when you can add factorial controls that separate structure vs decision rule.

Option B: "When consensus fails: calibration collapse makes majority voting brittle."

- Arc: Verifier is under-confident → consensus thresholds almost never pass → chunking + permissive decision rules outperform → focus on calibration-aware verification rather than compute per se.
- Best when you add a stronger calibration section (reliability diagrams + alternative aggregators).

Option C: "Trajectory drift as an agentic error-propagation failure mode."

- Arc: Feedback loops in agents → verifier rejection changes future state distribution → drift is a concrete error-propagation mechanism → propose mitigation by reducing sequential rejection events.
- Best when you strengthen drift definition + unbiased failure stats.

Abstract Outline (complete)

S1 (**Problem/setting**): State that web agents face irreversible commit actions (checkout) and world-model verifiers are used to screen them.

S2 (**Counterintuitive phenomenon**): State the observed non-monotonic regression: increasing verification *cycles* in sequential verify-and-retry can reduce task success.

S3 (**Mechanism + boundary conditions**): Attribute to rejection-induced re-planning that drifts trajectories, especially when the verifier is under-confident / has high false negatives.

S4 (**Method**): Define chunked allocation under fixed budget `B`: allocate into `M` cycles with `K=B/M` parallel rollouts per cycle; mention the decision rule class (consensus threshold vs single-rollout/permissive).

S5 (**Key evidence + scoped conclusion**): Report the main WebShop result (identify which condition achieves 21.50%), and state the scoped takeaway: minimizing sequential interventions is a robust default under poor calibration; sophisticated aggregation requires calibration.

Introduction Outline (complete; paragraph-by-paragraph)

P1 (Territory + stakes): Define irreversible commits in web agents and why verification is used; specify the *verifier-gated* setting (not full search/planning). Transition: "But the verification loop itself changes agent behavior."

P2 (Gap): Present the gap crisply: prior work improves world-model fidelity or uses rollouts for planning, but does not study **budget allocation across sequential verifier interventions** for irreversible commits. Transition: "This gap matters because sequential interventions can perturb the trajectory."

P3 (Phenomenon): State the non-monotonic regression with sequential verify-and-retry as cycles increase; include one concrete example narrative. Transition: "We hypothesize a specific mechanism."

P4 (Mechanism hypothesis): Define trajectory drift operationally (what counts as drift) and list 1–2 alternative hypotheses you will test (threshold effects, stochasticity). Transition: "We study allocation under a fixed budget."

P5 (Method overview): Introduce `(B,M,K)` and the decision rules; explicitly define what a cycle is and what happens on reject and on budget exhaustion. Transition: "We test this on WebShop under controlled budgets."

P6 (Results preview): Summarize the headline numbers with correct mapping to conditions; preview that consensus fails due to under-confidence and that smaller `M` wins in this setting. Transition: "We now review related work."

P7 (Contributions; scoped + testable): List 2–3 contributions as *testable statements* (e.g., "Under fixed `B`, smaller `M` improves success in WebShop; drift accounts for X% under an unbiased analysis; consensus fails under calibration collapse.").

Copy-ready rewrite targets (high-priority)

- **Page 1 - Abstract:** explicitly name that 21.50% is Condition C, and bound claims to the evaluated setting.
- **Page 1 - Intro P3:** replace "problem lies not in accuracy" with an interaction statement (verifier errors × sequential structure).
- **Page 3 - Verification Framework:** add Algorithm 1 (cycle consumption, reject behavior, budget exhaustion policy, τ selection).

Priority Revision Plan

P0 (Must fix before resubmission)

1. **Define the exact algorithm and policies (stop ambiguity).**
   - Add Algorithm 1 for verification; explicitly define cycle consumption, reject behavior, and budget exhaustion behavior.
   - Include the actor/verifier prompt interfaces and history truncation (Page 3 - Setup/Verification Framework).
2. **Run the confound-breaking experiment package (structure vs decision rule).**
   - Factorial: $M \in \{1,2,5,10\}$ × decision rule (single-rollout / consensus / any-positive) × τ sweep or tuned τ.
   - Add $B=0$ baseline.
   - Report CIs and key behavioral stats.
3. **Rewrite strongest causal claims to match what is isolated.**
   - Remove/qualify "not accuracy," "entirely structural," "dominant failure mode" unless the factorial + unbiased analyses confirm them.

P1 (High leverage improvements)

4. **Unbiased mechanism analysis for trajectory drift.**
   - Operational definition + symmetric conditional analysis + unconditional rates.
   - Run for C vs A (best method) in addition to B vs A.
5. **Calibration reporting upgrade.**
   - Reliability diagrams; specify ECE bins; cycle vs episode; stratify outcomes by $\hat{p}$.
   - If feasible, test a calibration/uncertainty baseline (e.g., confidence elicitation) [6][7].

P2 (Generalization + positioning)

6. **Generalization check on a second setting.**
   - Either a second benchmark with irreversible commits (small-scale) or a second verifier calibration regime.
7. **Related-work strengthening (novelty defensibility).**
   - Add a short "closest overlap" paragraph contrasting with world-model planning for irreversible actions [1][2] and test-time scaling for agents [4][5].

Page Coverage Audit

Page	Annotation Count	Coverage Status	Skip Reason
1	5	covered	—
2	5	covered	—

Page	Annotation Count	Coverage Status	Skip Reason
3	3	covered	—
4	2	covered	—
5	2	covered	—
6	2	covered	—

ASCII Diagram — Experiment Upgrade Plan

```
P0: Correctness + causal isolation
 (Algorithm 1) -> (B=0 baseline) -> (Factorial M × decision rule × tau)
                |
                v
P1: Mechanism + calibration
 (Unbiased drift labeling) + (Reliability diagrams + ECE details)
                |
                v
P2: Generalization
 (Second benchmark or second verifier) + (tight related-work positioning)
```

Experiment Inventory & Research Experiment Plan

Completed Experiment Inventory (as reported)

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E1	Compare sequential verify-and-retry vs chunked verification at fixed budget	WebShop episodes; `B=10`; 3 seeds; 200 tasks/seed (600 eps/condition). A: sequential (`K=1`, up to `M≈10`). B: chunked consensus (`M=1,K=10,τ=0.1`). C: chunked single-rollout (`M=1,K=10`, no aggregation). Actor: Gemini-2.5-Flash; Verifier: fine-tuned Qwen2.5-7B	Success rate; 95% bootstrap CI	A=0.86%, B=14.83%, C=21.50% (Page 4 - Main Results)	“Chunking improves success under same budget”; “consensus not helpful here”	Conditions differ in decision rule and τ; no `B=0` baseline; budget-exhaustion/auto-pass policy not ablated (Page 3 - Verification Framework/Experimental Conditions)

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E2	Isolate effect of verification cycle count `M` under fixed `B`	Vary `M∈{1,2,5,10}` with `B=10` and `K=B/M` (Page 4 - Cycle Reduction)	Success rate (figure)	M=1: 20.0%, M=2: 8.5%, M=5: 14.0%, M=10: 0.83%	“Fewer sequential cycles tends to help”	Confounded by `K` and consensus threshold interactions; CI per point not clearly reported in text; decision-rule details unclear (Page 4 - Cycle Reduction)
E3	Diagnose failure modes when chunked succeeds but sequential fails	Analyze episodes where B succeeds and A fails (n=81) (Page 5 - Failure Analysis)	Failure-category counts	Trajectory drift 60.5%, wrong product 27.2%, never reached checkout 12.3%	“Trajectory drift is a plausible mechanism”	Selection bias (conditioned subset); no symmetric/unconditional reporting; drift not operationally defined; not repeated for C vs A (Page 5 - Failure Analysis)
E4	Test whether gains come from stochastic diversity	Compare verifier decoding temp=0.7 vs 0.0 under configuration `M=2,K=5,τ=0.5` (Page 5 - Temperature Ablation)	Success rate + CI	Chunked advantage persists (and slightly increases) at temp=0	“Not primarily due to rollout diversity”	Not evaluated under the headline `M=1,K=10` configuration; actor stochasticity not ablated; large change in sequential baseline at temp=0 needs interpretation (Page 5 - Temperature Ablation)

Exp ID	Objective/Hypothesis	Setup (data/split/protocol/baselines)	Metrics	Main Outcome	Claim Supported	Current Limitation
E5	Explain why consensus fails via calibration statistics	Analyze Condition B verification cycles (311 cycles): distribution of `hat p`, Brier, ECE; empirical success when `hat p=0` (Page 6 - Calibration Analysis)	Calibration metrics	`hat p=0` for 98.4% cycles; when `hat p=0`, actual success 27.5%	"Verifier is severely under-confident; strict consensus rarely passes"	Need unit-of-analysis clarity (cycle vs episode), ECE binning details, and reliability diagram; conclusion "entirely structural" overreaches (Page 6 - Calibration Analysis)

Research-Theme Gap Diagnosis

- **New knowledge:** The paper's most valuable potential insight is that *sequential verifier rejections can create a harmful feedback loop* in irreversible-action settings. This is interesting, but currently not isolated from decision-rule artifacts.
- **Reproducibility / reusability:** Missing algorithm edge-case definitions (cycle consumption; budget exhaustion policy; τ selection; prompts/fine-tuning) limits reusability.
- **Potential to change practice:** High, if the paper can provide a **robust default rule** (e.g., small M + calibration-aware acceptance) with clear boundary conditions.

Proposed Research Experiments (P0/P1/P2)

Priority	Target Claim	Hypothesis	Minimal Design	Controls/Baselines	Metrics	Success Criterion	Estimated Cost/Time	Expected Paper-Quality Gain
P0	"Cycle reduction (small M) is the causal driver"	Holding decision rule fixed, increasing M decreases success due to re-planning drift	Factorial grid: $M \in \{1,2,5,10\}$; decision rule $\in$ {single-rollout, consensus, any-positive/max}; sweep or tune τ per K; keep B fixed	Add $B=0$ (no verification) and a "no auto-pass" variant	Success rate + CI; check out proposal count; block/pass rates	Clear monotone (or clearly explained non-monotone) trend with M under matched decision rules	Moderate: rerun ~ (grid size $\times$ 600 eps) or reduce to 200 eps for screening	Converts headline claim from suggestive to causal/isolation-grade

Priority	Target Claim	Hypothesis	Minimal Design	Controls/ Baselines	Metrics	Success Criterion	Estimated Cost/Time	Expected Paper-Quality Gain
P0	“Auto-pass policy does not create the gain”	The benefit persists without forcing checkout after `M` cycles	Compare budget-exhaustion policies: auto-pass vs abstain/terminate vs keep blocking	Same `M, K, τ` across policies	Success rate; #late checkouts	Chunked > sequential remains under multiple exhaustion policies	Low–moderate	Removes a major policy-artifact concern
P0	“Trajectory drift is dominant (unbiased)”	Drift frequency remains high under unbiased counting and for C vs A	Define drift operationally; compute unconditional failure taxonomy; symmetric pairwise sets (C succeeds/A fails; A succeeds/C fails)	Include C vs A; include B vs A	Drift rate; inter-annotator agreement (if manual)	Drift remains top category and is stable across comparisons	Moderate (requires logging/labeling)	Makes mechanism section credible and reusable
P1	“Consensus fails due to miscalibration (and can be fixed)”	Calibration-aware acceptance rules outperform strict majority under under-confidence	Add reliability diagram; compute ECE with stated bins; test a simple confidence elicitation / recalibration-inspired rule (e.g., ask verifier to output confidence) informed by [6][7]	Compare: consensus (various τ) vs any-positive/max vs confidence-elicited threshold	Calibration metrics + success	Either (i) calibrated rule improves consensus, or (ii) paper provides clear negative result with diagnostics	Moderate	Strengthens C3 from anecdote to calibrated-method story

Priority	Target Claim	Hypothesis	Minimal Design	Controls/ Baselines	Metrics	Success Criterion	Estimated Cost/Time	Expected Paper-Quality Gain
P1	“Robust across actor stochasticity”	Gains persist under different actor temperatures and/or fixed actor randomness	Sweep actor temp (e.g., 0, 0.5, 1.0) while keeping verifier fixed	Sequential vs chunked at each temp	Success; #re-browse steps after reject	Chunked advantage persists across actor stochasticity	Moderate	Clarifies whether drift is actor- or verifier-driven
P2	“Generalizes beyond one environment/verifier”	The qualitative effect (sequential harm) appears in another irreversible-action setting or another verifier calibration regime	Small-scale replication on another irreversible-action benchmark slice or with a second verifier; optionally connect to risk-aware simulation framing [8]	Report at least one additional setting	Success; drift frequency; calibration profile	Same directional findings	High	Upgrades the paper from case study to principle

Novelty Verification & Related-Work Matrix

9A) Contribution Novelty Verdict Board

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C1	Identify **trajectory drift** (rejection-induced re-browsing leading to worse checkout) as the root cause of non-monotonic regressions in sequential verify-and-retry verification for irreversible actions (Page 1 - Abstract/Introduction; Page 5 - Failure Analysis).	[4][8]	`partially_overlapping`	Prior agent test-time scaling/simulation work already highlights that sequential interventions can perturb long-horizon behavior and that irreversible actions require special care, but this paper's **specific drift mechanism + WebShop verification case study** appears to be a new empirical articulation.	Medium	Frame as a *verified-in-WebShop failure mode* and provide an operational definition + unbiased statistics before claiming "dominant."
C2	Propose **chunked budget allocation**: spend fixed budget `B` in fewer cycles `M` with more parallel rollouts per cycle (`K=B/M`) to prevent regressions (Page 1 - Abstract; Page 3 - Verification Framework; Page 4 - Main Results).	[4][5][8]	`partially_overlapping`	Compute allocation (parallel vs sequential) for agents is an active topic, and risk-aware simulation for irreversible actions exists. The novelty here is mainly the very simple "minimize sequential verifier interventions" rule applied to verifier-gated checkout.	Medium	Reposition as a *design rule for verifier-gated irreversible commits*; strengthen by isolating structure vs decision rule (factorial `M×` rule × `τ`) and by adding `B=0` baseline.

Claim ID	Author Contribution Claim	Key Evidence Papers [n]	Novelty Verdict Tag	Why	Confidence	Required Repositioning
C3	Show **consensus aggregation provides no benefit** because the verifier is severely under-confident (98.4% cycles with `hat p=0`), causing majority/threshold rules to degenerate (Page 6 - Calibration Analysis).	[6][7]	`partially_overlapping`	The broader point "self-consistency/aggregation can fail under miscalibration/top-skewed confidence" is well-known in confidence-estimation literature; the paper adds a **concrete instance** in world-model verification.	Medium-High	Present as a calibration-aware cautionary result; add reliability diagrams + clarify ECE/Brier computation to make this contribution technically solid.

9B) Related-Work Taxonomy Matrix

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Root	Verifier-gated irreversible actions for LLM agents (web/interactive)	—	Irreversible commits exist; test-time compute can be allocated	This paper focuses on *verification budget allocation* for checkout	Risk: needs crisp scope vs planning/search work
Branch	World models for web agents	—	LLMs can predict/simulate web transitions	This paper uses a world model mainly as a **checkout verifier**	Overlap risk: "simulate-before-commit" framing
Leaf	Model-based planning with LLM world models	[1]	Use world-model rollouts to choose actions	Planning/search vs accept/block verification; does not study `M`-cycle regressions	Moderate
Leaf	World-model-augmented web agent via transition abstraction	[2]	World model improves policy selection	Not centered on verification-budget allocation or non-monotonic regressions	Low–Moderate

Taxonomy Layer	Branch/Leaf	Representative Papers [n]	Common Assumptions	Difference vs This Paper	Novelty Risk Signal
Leaf	LLMs as text-based world models; agent utility incl. action verification	[3]	Action verification can improve agent utility	Does not analyze sequential verify-and-retry regressions nor chunking `M`	Moderate
Branch	Test-time compute allocation for agents	—	Extra compute can be spent in parallel or sequentially	This paper's "chunking" is a fixed-budget allocation rule	Overlap risk: compute allocation is not new; setting is specific
Leaf	Systematic test-time scaling for LLM agents	[4]	Parallel sampling + sequential revision + verifiers	Not specific to verifier-gated irreversible commits	Moderate
Leaf	Dynamic "when-to-plan" compute allocation for agents	[5]	Planning frequency matters in long-horizon tasks	Different lever (when to plan) vs (how to allocate verifier budget)	Moderate
Leaf	Risk-aware test-time simulation for irreversible actions	[8]	Simulate before acting; failure modes are costly	More complex simulation framework; this paper is a simpler heuristic	Moderate
Branch	Confidence estimation & calibration for LLM judges/verifiers	—	Confidence is measurable and useful for selective automation	This paper reports severe under-confidence and its effect on aggregation	Overlap risk: general phenomenon known
Leaf	Calibration of confidence for selective automation	[6]	Confidence should predict correctness; self-consistency may miscalibrate	Different domain; supports your calibration interpretation	Low
Leaf	Survey of confidence estimation & calibration in LLMs	[7]	Many calibration methods/metrics exist (ECE/Brier, etc.)	Background; suggests better reporting practices	Low

9C) Head-to-Head Comparison Matrix

Ref [n]	Problem/Setting	Method Core	Strongest Overlap Point	Clear Difference	Impact on Final Judgment
[1]	Web agents; irreversible actions limit backtracking	LLM world model for simulate-and-deliberate planning	“Simulate before commit” + irreversibility concern	Planning/search vs verifier-gated checkout budget allocation	Novelty depends on cleanly distinguishing verification-budget pathology from planning literature
[2]	WebArena/Mind2Web; irreversible mistakes	World-model-augmented policy selection via transition abstraction	World model used to avoid irreversible errors	Not a budget-allocation study; no non-monotonicity analysis	Supports relevance; does not invalidate novelty but raises positioning bar
[3]	Text-based environments; when world models help agents	Evaluation framework; shows action verification utility	Action verification as a use case	Does not study sequential verify-and-retry regressions or `M` vs `K` allocation	Suggests C2 is an extension; strengthen differentiation via controlled experiments
[4]	Test-time scaling for LLM agents	Parallel sampling, sequential revision, verifiers, merging	Parallel vs sequential compute trade offs; “when to reflect” matters	Not about irreversible commit verification loops	C2 is partially overlapping; paper should cite and position as a special-case principle
[5]	Long-horizon agentic tasks	Learn when to allocate compute for planning	Compute allocation in sequential decision-making	Different mechanism (dynamic planning) vs verifier gating	Reinforces that compute-allocation novelty is incremental; value is the specific irreversible-commit insight
[6]	Automated grading	Confidence calibration; self-consistency can be worse calibrated	Aggregation can hurt calibration; confidence top-skew	Different application	Backs C3 interpretation but reduces novelty; requires stronger calibration methodology in this paper
[7]	General LLM calibration	Survey of methods/metrics	Calibration metrics (ECE/Brier) and pitfalls	Background only	Helps authors improve calibration reporting and avoid overclaims

Ref [n]	Problem/Setting	Method Core	Strongest Overlap Point	Clear Difference	Impact on Final Judgment
[8]	Agentic settings with irreversible/costly actions	Risk-aware iterative simulation before execution	Irreversibility + test-time compute for safety	More complex simulator-based approach; this paper is a simpler allocation heuristic	Implies paper's value is as a low-cost baseline/design rule; needs generalization/controls

Contribution-level Novelty Conclusion. Based on retrieved related work, all three contributions appear **partially overlapping** rather than fully redundant. The paper's strongest residual novelty is the *empirical demonstration and mechanism articulation* that sequential verifier rejections can degrade irreversible-commit success via rejection-induced trajectory drift (C1), and the resulting simple mitigation "minimize sequential verifier interventions" under a fixed budget (C2). The "consensus fails under severe under-confidence" point (C3) aligns with known calibration findings [6][7] and should be positioned as a case study with stronger calibration reporting.

ASCII Diagram — Related-Work Taxonomy Tree (Layered)

```
Related Work Taxonomy (Root: verifier-gated irreversible actions + test-time
compute)
 ├── Branch A: World models for web agents
 │    ├── Leaf A1: Model-based planning with LLM world models [1]
 │    ├── Leaf A2: World-model-augmented policy selection / transition abstraction [
2]
 │    └── Leaf A3: LLMs as text-based world models; action verification utility [3]
 ├── Branch B: Test-time compute allocation for agents
 │    ├── Leaf B1: Test-time scaling strategies (parallel vs sequential, verifiers) [
4]
 │    ├── Leaf B2: Dynamic "when-to-plan" compute allocation [5]
 │    └── Leaf B3: Risk-aware test-time simulation for irreversible actions [8]
 └── Branch C: Confidence estimation & calibration for LLM judges/verifiers
      ├── Leaf C1: Empirical calibration for selective automation [6]
      └── Leaf C2: Calibration/uncertainty estimation survey & metrics [7]

This Paper (position): intersects A (verifier use of world model) + B (fixed-
budget allocation across
sequential cycles) + C (aggregation brittleness under under-confidence).
```

References

[1] Is Your LLM Secretly a World Model of the Internet? Model-Based Planning for Web Agents 2411.06559

[2] Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation 2410.13232

[3] From Word to World: Can Large Language Models be Implicit Text-based World Models? 2512.18832

[4] Scaling Test-time Compute for LLM Agents 2506.12928

[5] Learning When to Plan: Efficiently Allocating Test-Time Compute for LLM Agents 2509.03581

[6] When Can We Trust LLM Graders? Calibrating Confidence for Automated Assessment 2603.29559

[7] A Survey of Confidence Estimation and Calibration in Large Language Models 2311.08298

[8] ARTIS: Agentic Risk-Aware Test-Time Scaling via Iterative Simulation 2602.01709

Scores

Final Score: **5.5/10**

- Rationale (novelty + research value first): The paper targets an important deployment-relevant setting (irreversible commits) and surfaces a potentially high-impact empirical pathology (verification budget regressions). However, novelty is **incremental** relative to broader test-time compute allocation and world-model-for-web-agent work [1][2][4][8], and several validity-critical confounds/underspecifications (decision rule/threshold, budget-exhaustion policy, missing baselines/diagnostics) reduce confidence that the headline causal claim is established.

Post-Revision Target: **[7.0, 8.0]/10**

- If the authors add a precise algorithm spec, unconfounded factorial experiments, a `B=0` baseline, unbiased drift + stronger calibration reporting, and at least one small generalization check, the paper can become a strong design-principle note with clear practical guidance.

Appendix · Original Paper PDF

The following pages preserve source layout and include review annotations when available.

The appendix below attaches source paper pages and overlays review highlights for ISSUE / SUGGESTION / EVIDENCE objects.

Callout boxes are placed near page margins to avoid covering core content and remain within page boundaries.

End of report · DeepScientist Review Export

CHUNKED BUDGET ALLOCATION PREVENTS NON-MONOTONIC REGRESSIONS IN WORLD-MODEL VERIFICATION

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ABSTRACT

World models that predict environment dynamics can serve as pre-execution verifiers for agents facing irreversible actions. However, sequential verify-and-retry—where rejected actions trigger agent re-planning—can paradoxically *reduce* task success as verification budget increases. We identify *trajectory drift* as the root cause: when verification rejects a checkout (often a false negative), the agent re-browses and frequently selects a worse product than originally found. We propose *chunked budget allocation*: instead of spreading verification budget across many sequential cycles, spend it in fewer cycles with more parallel rollouts per cycle. On WebShop, chunked verification ($M = 1$ cycle) achieves 21.50% task success versus 0.86% for sequential verify-and-retry ($M = 10$ cycles) at the same budget. Surprisingly, consensus aggregation across parallel rollouts provides no benefit over single-rollout acceptance, as the world model's poor calibration (predicting failure for 98.4% of cycles) causes consensus to degenerate. Our results demonstrate that budget allocation structure matters more than aggregation sophistication when world models are poorly calibrated.

WARNING: This paper was generated by an automated research system. The code is publicly available.[1]

1 INTRODUCTION

World models that predict environment dynamics have emerged as a powerful tool for agent decision-making (Ha & Schmidhuber, 2018; Schrittwieser et al., 2019). For agents operating in environments with irreversible actions—such as submitting purchases, sending emails, or deleting files—world models can serve as pre-execution verifiers: before committing to a high-stakes action, the agent queries the world model to predict whether the action will succeed. Recent work has demonstrated this approach for web agents, using LLM-based world models to verify actions before execution (Li et al., 2025; Gu et al., 2025).

However, a counterintuitive failure mode has been observed: increasing the verification budget (the number of world-model queries allowed) can *reduce* task success rates (Li et al., 2025). This non-monotonic behavior occurs because sequential verify-and-retry creates a problematic feedback loop. When verification rejects a checkout (potentially a false negative), the agent re-browses to find another product. During re-browsing, the agent often selects a *worse* product than the one originally found—a phenomenon we call *trajectory drift*. More verification cycles mean more opportunities for trajectory drift.

The key insight is that the problem lies not in verification accuracy but in the *sequential structure* of verify-and-retry. Each rejection triggers re-browsing, compounding errors across cycles. This suggests a simple solution: instead of spreading the verification budget across many sequential cycles, spend it in fewer cycles with more parallel rollouts per cycle.

We propose *chunked budget allocation*: given a fixed verification budget B, execute M cycles with $K = B/M$ parallel world-model rollouts per cycle. By setting $M = 1$, we spend the

[1] https://gitlab.com/fars-a/worldmodel-consensus

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Index ID: #P01-A04
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Annotation Continuation · Source Page 1

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#P01-A01 · SUGGESTION · MAJOR · #R001 · Part 1

Page 1 - Abstract

Problem (claim/evidence alignment): The abstract mixes three distinct ideas—(i) ***budget allocation across sequential cycles*** (M vs. K), (ii) ***aggregation rule*** (consensus vs. single rollout), and (iii) a very large headline gain (21.50% vs 0.86%)—but it is not explicit that the 21.50% result corresponds to the **no-aggregation** chunked variant (Condition C) rather than the consensus method (Condition B). This can mislead readers about what the proposed method actually is. It also implicitly suggests broad "verification budget increases hurt" generality, while the evidence in this paper is limited to one environment/model pair.

Why it matters: For a short paper, the abstract is the main contract with the reader. If the method/result mapping is ambiguous, reviewers may interpret the contribution as "we propose consensus chunking" but the strongest result comes from a different decision rule. Also, without a *no-verification* baseline (B=0), the magnitude of improvement is hard to interpret: chunking might mainly behave like "stop rejecting repeatedly" rather than adding useful verification.

Actionable fix (content + defensible scope):
- Explicitly name which chunked variant achieves 21.50% (e.g., "M=1 with single-rollout acceptance").
- Add one short phrase bounding scope: "in this WebShop setting with a poorly calibrated verifier".
- If space allows, report (or promise in the paper) a no-verification baseline to contextualize whether chunking beats simply not verifying.

Mentor Revised Version:
World-model verifiers are often used to screen irreversible actions (e.g., checkout) in web agents. We show that a standard sequential verify-and-retry loop can reduce WebShop task success as more verification cycles are allowed, because rejected checkouts trigger re-browsing that drifts the trajectory away from previously good candidates. We study budget allocation under a fixed number of world-model calls B by distributing calls across M verification cycles with $K=B/M$ parallel rollouts per cycle; smaller M reduces verification-induced re-planning. On WebShop (Gemini-2.5-Flash actor; fine-tuned Qwen2.5-7B verifier) allocating the full budget to a single cycle ($M=1$, $K=10$) achieves 21.50% success with single-rollout acceptance, versus 0.86% for the sequential baseline ($Mapprox10$, $K=1$) at the same $B=10$. In this setting, consensus aggregation does not help because the verifier is extremely under-confident (98.4% of cycles yield $hat p=0$), suggesting allocation structure can dominate aggregation when verifiers are poorly calibrated.

References:
[4] Scaling Test-time Compute for LLM Agents 2506.12928

[9] ARTIS: Agentic Risk-Aware Test-Time Scaling via Iterative Simulation 2602.01709

#P01-A02 · SUGGESTION · MAJOR · #R002 · Part 1 · CONT.

Page 1 - Introduction (Paragraph 1: motivation + prior work)

Problem: This paragraph sets up "world models as pre-execution verifiers" and cites prior web-agent world-model work, but it blurs two different uses of world models: (i) model-based planning/search (simulate transitions to choose actions) and (ii) *verification* of a proposed irreversible action. Because the rest of the paper is about verification budget allocation, readers need a sharper statement of the *specific gap*: not "world models exist," but "world-model verification + retry can regress with more budget in irreversible settings." Also, "world model" is used broadly; later you use a fine-tuned Qwen model predicting next observation + success indicator, which is a particular operational definition.

Why it matters: Novelty and contribution are easiest to reject when the problem statement does not crisply distinguish the paper's setting from adjacent model-based planning literature (e.g., using world models for deliberation over action outcomes). This is especially important because prior work explicitly discusses irreversible actions on the web and uses world-model simulation for decision-making.

Actionable fix:
- Add a one-sentence operational definition: in this paper, the verifier predicts a binary task-success outcome for a candidate irreversible action given the current trajectory history.
- Add a one-sentence gap statement that foreshadows non-monotonic regressions under sequential verify-and-retry.
- Add a short clause contrasting with model-based planning (you are not proposing search/planning; you are analyzing budget allocation for a verifier).

Mentor Revised Version:
World models can help web agents avoid costly irreversible mistakes by *predicting the outcome of a proposed action before execution*. In this paper, we focus on a verifier-style world model that, given the current trajectory history, predicts whether a candidate irreversible action (here: checkout) will satisfy the user intent. This differs from model-based planning methods that use simulated rollouts to *choose* actions via search:

Annotation Continuation · Source Page 1 (cont. 2)

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#P01-A02 · SUGGESTION · MAJOR · #R002 · Part 2

our setting assumes the agent already proposes a checkout and the world model is queried only to accept or block it. A key open question is how to use a fixed verification budget effectively in such irreversible environments, since sequential "verify-and-retry" loops can change the agent's future trajectory after each rejection.

References:
[1] Is Your LLM Secretly a World Model of the Internet? Model-Based Planning for Web Agents 2411.06559

[2] Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation 2410.13232

#P01-A03 · SUGGESTION · MAJOR · #R003 · Part 1

Page 1 - Introduction (Paragraph 2: non-monotonicity + trajectory drift)

Problem: This paragraph asserts a strong causal story: more verification budget → more sequential cycles → more false-negative rejections → re-browsing → worse product → lower success ("trajectory drift"). The paper later provides a failure-pattern table, but the current paragraph reads like a complete explanation without acknowledging alternative mechanisms (e.g., verifier thresholding effects, acting-agent exploration temperature, or budget exhaustion rule "auto-pass after M cycles" changing behavior). Also, "often a false negative" is plausible but not quantified here; readers need an explicit statement of what evidence will be shown.

Why it matters: The paper's value hinges on identifying the *right* mechanism. If trajectory drift is only one contributor, or if the observed effect is partially an artifact of decision thresholds or policy changes after budget exhaustion, reviewers will question the generality of the recommendation "set M=1."

Actionable fix:
· In this paragraph, downgrade causal language to "suggests/is consistent with" and preview the specific evidence you will provide (e.g., the fraction of failures attributable to re-browsing and an estimate of verifier false negative rate conditioned on ground-truth success).
· Add one sentence naming at least one alternative hypothesis you will rule out later (e.g., threshold/consensus interaction, acting-agent stochasticity).

Mentor Revised Version:
Prior work reports a counterintuitive failure mode: allowing more verifier queries can reduce end-to-end task success in WebShop-style settings. We hypothesize that this regression is largely driven by *trajectory drift*: when a verifier blocks an initially good checkout (potentially due to false negatives), the agent re-enters browsing and may settle on an inferior product, so additional verify-and-retry cycles create more opportunities to drift away from an earlier good candidate. In Section 4, we test this hypothesis by (i) comparing different allocations of the same verification budget across sequential cycles, and (ii) categorizing failures to measure how often re-browsing after a rejection precedes the eventual mistake. We also examine alternative explanations such as threshold/aggregation effects and decoding stochasticity.

References:
[4] Scaling Test-time Compute for LLM Agents 2506.12928

[5] DynScaling: Efficient Verifier-free Inference Scaling via Dynamic and Integrated Sampling 2506.16043

#P01-A04 · SUGGESTION · MAJOR · #R004 · Part 1 · CONT.

Page 1 - Introduction (Paragraph 3: "key insight")

Problem: The paragraph states "the problem lies not in verification accuracy, but in the sequential structure," which is too absolute given later evidence that the verifier is *poorly calibrated* and has a high false-negative tendency. A more defensible claim is conditional: even with imperfect verifiers, sequential rejection-induced re-planning can dominate outcomes; structural choices interact with verifier error rates.

Why it matters: Reviewers will push back on causal overreach ("not accuracy") because the paper itself argues consensus fails due to calibration (an accuracy/uncertainty quality issue). The scientific contribution is stronger if you frame an interaction: structure × calibration × re-planning dynamics.

Actionable fix:
· Rephrase to: "the regression is driven by the interaction between verifier errors and the sequential verify-and-retry structure."
· Add one sentence connecting to test-time compute allocation in agents: you are studying *how to spend a fixed budget* (parallel vs sequential) for

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#P01-A04 · SUGGESTION · MAJOR · #R004 · Part 2

safety-critical/irreversible steps, which relates to broader test-time scaling strategies.

Mentor Revised Version:
A key observation is that non-monotonic regressions can arise from an *interaction* between verifier errors (e.g., under-confident or miscalibrated predictions) and the sequential verify-and-retry structure: each rejection triggers re-browsing, so even occasional false negatives can push the agent onto worse trajectories over time. This motivates viewing verification as a test-time compute allocation problem in an agentic setting: under a fixed world-model query budget, should we spend compute sequentially across many retries, or concentrate it into fewer decision points with more parallel rollouts?

References:
[4] Scaling Test-time Compute for LLM Agents 2506.12928

[5] Uncertainty-Aware Budget Allocation for Adaptive Test-Time Reasoning 2605.26849

#P01-A05 · SUGGESTION · MAJOR · #R005 · Part 1

Page 1 - Introduction (Paragraph 4: method definition + contributions lead-in)

Problem: The definition "allocate budget B across M cycles with K=B/M parallel rollouts per cycle" is clear, but the paragraph omits key operational details needed to interpret the rest of the paper:
· What exactly is a "cycle" (is it "each time the agent proposes checkout")?
· When a checkout is blocked, does the agent continue with the same policy and same context window? Is there any memory of the previously found good product?
· How is the decision made in the chunked case (consensus threshold vs first rollout), and how is tau chosen?

These omissions make the upcoming comparisons feel like black-box configurations rather than a well-defined algorithmic family.

Why it matters: The core scientific claim is that *allocation structure* causes the performance change. To make this causal story credible and reproducible, the reader needs a precise algorithm definition, especially because rejection changes the agent's subsequent observation distribution.

Actionable fix:
· Add a short formal definition: "A cycle is triggered whenever the actor proposes checkout; the verifier returns accept/block; if block, the actor continues from the current environment state for up to 50 steps."
· Add explicit decision rules and hyperparameters (tau for consensus, auto-pass rule when budget exhausted) in this paragraph or immediately after.
· Tighten the contribution list to match what is actually tested (and avoid overclaiming that the guidance is universal).

Mentor Revised Version:
We study a family of verifier policies under a fixed budget of B world-model calls per episode. Each time the actor proposes an irreversible action (checkout), we run one *verification cycle*: we query the world model K times from the current trajectory state, aggregate these rollouts into an accept/block decision (e.g., by thresholding $hat p$ or using a single rollout), and if blocked, the actor continues browsing in the real environment until the next checkout proposal. We allocate the same total budget across M cycles with $K=B/M$ rollouts per cycle. Our hypothesis is that smaller M reduces rejection-induced re-planning and thus mitigates trajectory drift.

References:
[1] Is Your LLM Secretly a World Model of the Internet? Model-Based Planning for Web Agents 2411.06559

[9] ARTIS: Agentic Risk-Aware Test-Time Scaling via Iterative Simulation 2602.01709

entire budget in one parallel chunk, eliminating the sequential cycles that trigger trajectory drift. Our contributions are:

- We identify *trajectory drift* as the dominant failure mode in sequential verify-and-retry, accounting for 60.5% of failures when agents re-browse after verification rejection.
- We demonstrate that chunked budget allocation achieves a 25× improvement in task success rate (21.50% vs. 0.86%) by eliminating sequential verification cycles.
- We show that consensus aggregation across parallel rollouts provides no benefit over single-rollout acceptance, as the world model's poor calibration (predicting $\hat{p} = 0$ for 98.4% of cycles) causes consensus to degenerate.
- We provide practical guidance: setting $M = 1$ (spending all budget in one parallel chunk) consistently outperforms finer-grained allocations, with performance degrading monotonically as M increases.

2 Related Work

Web Agents. The development of autonomous web agents has ... alistic benchmarks. WebShop (Yao et al., 2022) introduced a ... where agents must navigate product search and purchase workflows ... extended this to self-hosted websites with more complex, multi-step tasks. Mind2Web (Deng et al., 2023) provided a large-scale dataset of real-world web interactions across diverse domains, while WebVoyager (He et al., 2024) demonstrated end-to-end web navigation using multimodal models. We use WebShop as our testbed due to its well-defined irreversible ... evaluation protocol.

World Models for Agents. World models that predict environment ... for agent planning. Classical approaches (Ha & Schmidhuber, 2018; Schrittwieser et al., 2019) learn latent dynamics models for game environments. Recent work has explored LLMs as implicit world models for web agents: WebDreamer (Gu et al., 2025) uses LLMs to simulate web page transitions for model-based planning, Word2World (Li et al., 2025) investigates whether LLMs can serve as text-based world simulators, and DynaWeb (Ding et al., 2026) applies model-based reinforcement learning to web navigation. Chae et al. (2024) train world models to predict HTML state transitions. Our work is complementary: rather than improving world model ... a fixed verification budget effectively.

Test-Time Scaling. Scaling compute at test time has emerged as ... ing LLM performance. Self-consistency (Wang et al., 2022) samples multiple reasoning paths and aggregates via majority voting. Recent work extends this to agentic settings: Lee et al. (2026) study test-time scaling for web agents, and Zhu et al. (2025) investigate compute allocation strategies for LLM agents. Our findings reveal a surprising limitation: for verification of irreversible actions, the allocation structure (chunked vs. sequential) matters more than the aggregation rule (consensus vs. first-rollout).

Error Recovery in Agents. Agents inevitably make mistakes, ... anisms. Reflexion (Shinn et al., 2023) enables agents to learn from verbal feedback on failed attempts. WebRollback (Zhang et al., 2025) introduces explicit rollback mechanisms for web agents to recover from errors. However, these approaches assume that retry improves outcomes. We show that for irreversible actions, sequential retry can cause trajectory drift, where rejected checkouts trigger re-browsing that leads to worse product selections.

3 Method

3.1 Problem Setup

We study world-model verification for irreversible actions in the WebShop environment (Yao et al., 2022), a text-based e-commerce benchmark where an agent must navigate product search and complete purchases. The agent receives text observations describing search results and product pages,

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#P02-A01 · SUGGESTION · MINOR · #R006 · Part 1

Page 2 - Related Work (Web Agents paragraph)

Problem: The paragraph enumerates benchmarks (WebShop/WebArena/Mind2Web/WebVoyager) but does not articulate the *comparison axis* that matters for this paper (irreversible actions + evaluation protocol + whether retries are allowed). As written, it feels like a generic benchmark survey rather than positioning for a verification-budget-allocation study.

Why it matters: Your paper's thesis is about verification for irreversible actions; the related work should teach the reader why WebShop is the right "minimal" environment for isolating this phenomenon, and what the missing evidence is for broader generality (e.g., real websites, multimodal, longer horizons).

Actionable fix:
- Add 1–2 sentences that define the axis: "irreversible commit action with episode termination" and "rule-based success metric".
- Briefly justify why WebShop, despite being simulated, is appropriate for isolating verify-and-retry regressions.

Mentor Revised Version:
Web-agent benchmarks differ not only in UI realism but also in whether they contain *irreversible commit actions* that terminate an episode and make backtracking infeasible. WebShop provides a controlled setting with an explicit irreversible checkout step and a standardized success metric, which makes it suitable for isolating how verifier decisions interact with sequential retries. In contrast, benchmarks such as WebArena, Mind2Web, and WebVoyager increase website realism and modality, but also introduce additional confounders (dynamic pages, multimodal grounding, longer horizons) that we leave to future generalization tests.

References:
[8] AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories 2504.08942

#P02-A02 · SUGGESTION · MAJOR · #R007 · Part 1

Page 2 - Related Work (World Models for Agents paragraph)

Problem: This paragraph correctly cites prior LLM world-model work for web agents, but the positioning is still somewhat generic ("complementary"). Given retrieval evidence, prior work already emphasizes irreversible actions and using world models to simulate outcomes before committing (model-based planning). The manuscript's novelty is *not* "world models for web agents," but "how to allocate a fixed verification/query budget across sequential vs parallel verification cycles to avoid regression." That core differentiation is not made explicit.

Why it matters: Without an explicit head-to-head overlap statement, novelty risk is high: a reviewer can argue that your setting is a special case of prior "simulate before commit" approaches (e.g., pre-execution deliberation), and that the paper is mainly an ablation of K vs M with a particular verifier.

Actionable fix:
- Add a short paragraph at the end of this subsection that states: (i) what prior work optimizes (world-model fidelity/planning/search), (ii) what you hold fixed (verifier model), and (iii) what you study (budget allocation + sequential feedback effects).
- Cite the closest overlaps explicitly and state the new failure mode + design rule.

Mentor Revised Version:
Prior work leverages LLM-based world models to improve web agents primarily by increasing world-model fidelity or using simulated rollouts for planning and policy selection. In contrast, we hold the verifier model fixed and study a different question: under a fixed per-episode query budget, how should verification compute be allocated across sequential verify-and-retry cycles versus parallel rollouts at a single decision point? We show that, for irreversible commit actions, sequential rejection-induced re-planning can cause trajectory drift that overwhelms the benefits of additional verification.

References:
[1] Is Your LLM Secretly a World Model of the Internet? Model-Based Planning for Web Agents 2411.06559

[2] Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation 2410.13232

[3] From Word to World: Can Large Language Models be Implicit Text-based World Models? 2512.18832

Annotation Continuation · Source Page 2 (cont. 2)

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#P02-A03 · SUGGESTION · MAJOR · #R008 · Part 1

Page 2 - Related Work (Test-Time Scaling paragraph)

Problem: The paragraph claims "allocation structure matters more than aggregation rule," but it does not connect to existing test-time scaling findings that (a) sequential reflection can hurt if applied indiscriminately, and (b) compute allocation should be adaptive to uncertainty. Also, your "consensus fails" explanation is really about confidence estimation/calibration collapse (mass at $hat p=0$), which has parallels in LLM confidence estimation and LLM-as-a-judge calibration literature.

Why it matters: This is where you can make the paper feel like more than a one-off WebShop ablation. By grounding your claim in known test-time scaling/control phenomena, you can position chunking as an *agentic compute allocation* principle for irreversible commits.

Actionable fix:
· Add one sentence acknowledging that prior test-time scaling often finds positive effects from parallel sampling, but sequential strategies can degrade long-horizon performance when they repeatedly perturb the trajectory.
· Add one sentence situating your consensus failure as a calibration issue and referencing relevant calibration work.

Mentor Revised Version:
Test-time scaling for LLM agents spans both parallel sampling and sequential revision/reflection. While more compute often helps, sequential interventions can also degrade long-horizon trajectories when they repeatedly alter the agent's state distribution after each rejection or revision. Our results highlight a complementary axis for agentic settings with irreversible actions: under a fixed query budget, concentrating verification compute into fewer decision points can reduce harmful sequential feedback loops. We further observe that naive consensus can fail when verifier confidence is severely miscalibrated, aligning with broader findings on confidence estimation and calibration for LLM-based judges.

References:
[4] Scaling Test-time Compute for LLM Agents 2506.12928

[6] When Can We Trust LLM Graders? Calibrating Confidence for Automated Assessment 2603.29559

[7] A Survey of Confidence Estimation and Calibration in Large Language Models 2311.08298

#P02-A04 · SUGGESTION · MINOR · #R009 · Part 1

Page 2 - Related Work (Error Recovery in Agents paragraph)

Problem: The paragraph states "these approaches assume retry improves outcomes," which is not always accurate: rollback/backtracking methods often explicitly manage recovery. The real distinction for your work is narrower: even with recovery mechanisms, repeated retry that changes the browsing trajectory can degrade final selection under irreversible commits, especially when the verifier has a high false-negative rate.

Why it matters: Overstating assumptions about prior work can backfire in review. A precise contrast improves credibility and helps define the boundary conditions of "trajectory drift."

Actionable fix:
· Replace "assume retry improves outcomes" with a more bounded statement: "typically aim to make retries effective via reflection/rollback, but do not study the budget-allocation-induced regression mechanism in verifier-gated irreversible commits."

Mentor Revised Version:
Prior error-recovery mechanisms (e.g., reflection, rollback/backtracking) aim to make retries more effective by using feedback or state restoration. Our focus is different: in verifier-gated irreversible commits, repeated *re-planning after a blocked commit* can change what the agent subsequently browses and selects, so additional retry opportunities can systematically increase the chance of drifting away from an initially good candidate when the verifier is under-confident.

References:
[2] Web Agents with World Models: Learning and Leveraging Environment Dynamics in Web Navigation 2410.13232

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#P02-A05 · SUGGESTION · MAJOR · #R010 · Part 1

Page 2 - Method (3.1 Problem Setup)

Problem: The environment and irreversibility are described well, but key protocol details needed to reproduce and interpret the results are missing here:

- When does the agent decide to propose “checkout” (is it a learned/prompted action each step, or only from a product page)?
- Is WebShop deterministic given an action sequence (important for reasoning about retries and drift)?
- How exactly is “purchase matches user intent” computed (rule-based string match? WebShop’s official evaluator details)?
- What is the baseline when no verification is used (B=0), and is it comparable to your conditions?

Why it matters: Your central claim is about *decision dynamics* under rejection; small protocol differences (termination rules, allowed actions, evaluator) can change whether sequential retries help or hurt. Missing setup details also makes the reported absolute success rates hard to contextualize.

Actionable fix:

- Add 3–5 sentences here defining: action space constraints around checkout; deterministic vs stochastic environment aspects; the exact success evaluation; and a “no verification” baseline used for sanity.

Mentor Revised Version:
We study verifier-gated checkout in WebShop, where the agent may execute a checkout action at any step after navigating to a specific product page. Checkout is irreversible: once executed, the episode terminates and success is scored by the WebShop official evaluator based on whether the purchased item satisfies the task constraints (e.g., attributes and price). Throughout this paper we report success rate under a fixed maximum of 50 interaction steps and compare against a no-verification baseline ($B=0$) to isolate whether verification is beneficial beyond simply reducing retries.

References:
[8] AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories 2504.08942

and takes actions such as searching, clicking products, and checking out. Checkout is an *irreversible action*: once executed, the episode terminates regardless of whether the purchase matches the user's intent.

Our setup uses Gemini-2.5-Flash as the acting agent and a fine-tuned Qwen2.5-7B model as the world model (Li et al., 2025). Given the current trajectory history h and a candidate action a, the world model predicts the next observation s' and a binary success indicator $R' \in \{0, 1\}$. For checkout verification, R' indicates whether the purchase would satisfy the user's intent.

Index ID: #P03-A01
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3.2 Verification Framework

Given a verification budget B (maximum world-model calls per episode), we consider how to allocate this budget across M verification cycles, with $K = B/M$ parallel rollouts per cycle. When the agent proposes checkout, we generate K independent stochastic world-model predictions from the current state and compute the consensus estimate:

$$\hat{p} = \frac{1}{K} \sum_{k=1}^{K} \mathbb{1}[R'_k = 1] \tag{1}$$

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The checkout proceeds if $\hat{p} \geq \tau$ (threshold), otherwise it is blocked and the agent continues browsing. After M cycles, any subsequent checkout attempt automatically passes (budget exhausted).

3.3 Experimental Conditions

We compare three verification strategies at the same budget $B = 10$:

Condition A (Sequential Verify-and-Retry). The standard approach from prior work (Li et al., 2025): each checkout proposal triggers a single world-model query ($K = 1$). If $R' = 1$, checkout proceeds; if $R' = 0$, checkout is blocked and the agent continues. This allows up to $M \approx 10$ verification cycles.

Condition B (Chunked Consensus). Our proposed approach: spend the entire budget in $M = 1$ cycle with $K = 10$ parallel rollouts. Checkout proceeds if $\hat{p} \geq \tau$ (we use $\tau = 0.1$). This eliminates sequential verification cycles that could trigger trajectory drift.

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Condition C (Chunked No-Aggregation). A control condition with the same structure as B ($M = 1$, $K = 10$) but without consensus aggregation: only the first rollout's R' determines the decision. This isolates whether gains come from reducing cycles (shared by B and C) or from consensus aggregation (only in B).

Figure 1 illustrates the key structural difference between these approaches.

4 Experiments

4.1 Experimental Setup

We evaluate on WebShop (Yao et al., 2022), a text-based e-commerce benchmark with 500 test tasks. Each task specifies a user intent (e.g., "find a red cotton t-shirt under $30"), and the agent must navigate search results, browse product pages, and complete a purchase that matches the intent. We use 200 randomly sampled tasks per seed across 3 seeds (600 total episodes per condition). The primary metric is **task success rate**: the percentage of episodes where the purchased product satisfies the user's intent.

The acting agent is Gemini-2.5-Flash (temperature 1.0) and the world model is a fine-tuned Qwen2.5-7B (Li et al., 2025) served via vLLM (temperature 0.7, top_p 0.9). Each episode has a maximum of 50 steps. All conditions use the same verification budget $B = 10$. We report 95% bootstrap confidence intervals computed over the 600 episodes.

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#P03-A01 · SUGGESTION · MAJOR · #R011 · Part 1

Page 3 - Method (end of 3.1 Problem Setup)

Problem: The paragraph specifies the acting agent (Gemini-2.5-Flash) and verifier (fine-tuned Qwen2.5-7B) but omits details necessary to reproduce or even interpret the reported failure modes:
· Acting agent prompt template and whether it is allowed to see verifier outputs.
· Context window/history truncation policy for h (important for "trajectory drift").
· World model fine-tuning data source/size/objective: is it trained on WebShop trajectories? Does it predict both s' and R' jointly? How is R' supervised?
· Sampling parameters are given later, but not whether R' is sampled or deterministic.
Also, there is a notation/extraction issue: "candidate action ^{a,}" is unclear and looks like a formatting bug.

Why it matters: The core claim depends on false negatives and calibration; these are properties of the verifier model and its training. Without training/evaluation details, readers cannot know whether the phenomenon is general or an artifact of a specific fine-tuning recipe.

Actionable fix:
· Add a short "Implementation details" paragraph (or appendix) describing the actor prompt, max context length and truncation, and verifier fine-tuning procedure.
· Fix the action notation and consistently denote action as a.

Mentor Revised Version:
Our setup uses Gemini-2.5-Flash as the acting policy and a fine-tuned Qwen2.5-7B model as the world-model verifier. Given the trajectory history h and a candidate checkout action a, the verifier produces a predicted next observation s' and a binary success indicator $R'in{0,1}$ indicating whether executing checkout would satisfy the user intent. For reproducibility, we provide (i) the full actor prompt and whether verifier feedback is included in the actor context, (ii) the history truncation policy used to construct h, and (iii) the verifier fine-tuning dataset and objective used to supervise R'.

References:
[3] From Word to World: Can Large Language Models be Implicit Text-based World Models? 2512.18832

#P03-A02 · ISSUE · MAJOR · #R012 · Part 1 · CONT.

Page 3 - Method (3.2 Verification Framework)

Problem: The framework defines $hat p$ and a threshold rule, but several details are underspecified and can materially change outcomes:
1) "After M cycles, any subsequent checkout attempt automatically passes" is a strong policy choice that may inflate success (or induce failures) compared to alternatives (e.g., stop the episode, or keep blocking). It also couples performance to when the agent first proposes checkout.
2) It is unclear whether a "cycle" is consumed only when the agent proposes checkout, and whether multiple checkout proposals can occur (likely yes).
3) How is tau selected? In later experiments tau varies (0.1 vs 0.5), and the observed non-monotonicity partially depends on this interaction. Without a principled selection rule (calibration set, fixed across M, etc.), the comparison may be unfair.

Why it matters: Your main claims are about budget allocation structure. If auto-pass and tau choices change, the claimed mechanism could partially be a policy artifact rather than an inherent structural effect. This is a validity/reproducibility risk.

Actionable fix:
· Add an explicit "Algorithm 1" pseudocode (or boxed procedure) defining: when cycles are consumed, what happens on reject, what happens on budget exhausted, and which decision rule is used in each condition.
· Add an acceptance-criteria section: define tau selection protocol (e.g., fixed global value chosen on a dev set) and keep it constant across compared methods unless explicitly studied.

Mentor Revised Version:
We define a verification cycle as the event when the actor proposes checkout. In cycle $mle M$, we query the world model $K=B/M$ times from the current trajectory state, compute $hat p=(1/K)sum_k mathbb{1}[R_k'=1]$, and execute checkout iff $hat pge tau$; otherwise checkout is blocked and the actor continues browsing. If the actor proposes checkout after M consumed cycles, we either (a) execute checkout unconditionally (auto-pass; used in this study) or (b) terminate/abstain; we report both variants in ablations. Unless otherwise stated, tau is selected once on a held-out validation set and fixed across all M.

References:

Annotation Continuation · Source Page 3 (cont. 2)

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#P03-A02 · ISSUE · MAJOR · #R012 · Part 2

[6] When Can We Trust LLM Graders? Calibrating Confidence for Automated Assessment 2603.29559

[7] A Survey of Confidence Estimation and Calibration in Large Language Models 2311.08298

#P03-A03 · ISSUE · MAJOR · #R013 · Part 1

Page 3 - Method (3.3 Experimental Conditions)

Problem: The three conditions are meant to isolate structure vs aggregation, but the current definition creates comparability/interpretation issues:
· Condition B uses consensus with $tau=0.1$, while later mechanism analysis discusses $tau=0.5$ interactions and temperature ablation uses $M=2,K=5,tau=0.5$. This makes it hard to know whether B underperforms C because of aggregation per se, or because the chosen threshold is suboptimal under miscalibration.
· Condition C "first-rollout acceptance" is not just "no aggregation"; it changes the effective decision rule in a way that can systematically increase pass probability when the model is under-confident (because consensus averages many zeros). This is *the* point of your finding, but it should be framed as: "consensus is brittle under extreme under-confidence; single-sample decision can be less brittle."

Why it matters: The strongest causal claims ("consensus provides no benefit") require a fair comparison across thresholds/decision rules. Otherwise reviewers can argue the conclusion is an artifact of one chosen tau.

Actionable fix:
· Add a controlled sweep: for fixed $M=1,K=10$, evaluate multiple tau values for consensus (including very low thresholds) and compare to alternative aggregators (max, softmax, any-positive) that are natural under under-confidence.
· Clarify the hypothesis test: (H1) structure effect (M) vs (H2) aggregation/decision-rule effect given calibration.

Mentor Revised Version:
We compare (A) sequential verify-and-retry ($Mapprox B, K=1$), (B) chunked parallel verification with consensus thresholding, and (C) chunked parallel verification with a single-sample decision. To separate allocation effects from decision-rule effects, we additionally sweep the consensus threshold tau for (B) under the same $M=1,K=10$ setting and report which tau maximizes performance. We interpret any residual gap between (B) and (C) as evidence that, under severe under-confidence, majority-style aggregation can be brittle.

References:
[6] When Can We Trust LLM Graders? Calibrating Confidence for Automated Assessment 2603.29559

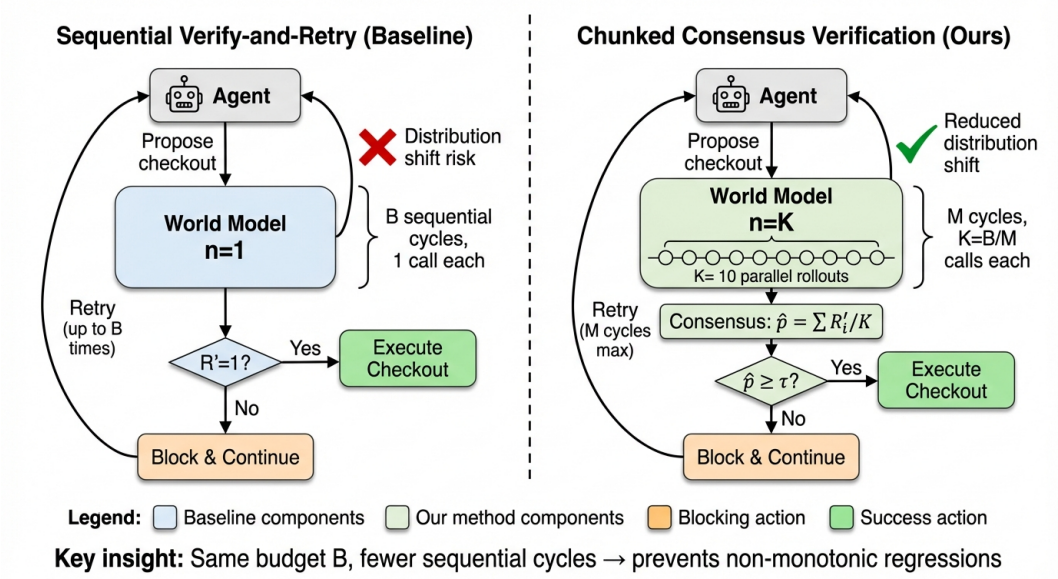


Figure 1: Comparison of verification strategies for irreversible actions. **Left**: Sequential verify-and-retry uses $K = 1$ rollout per cycle, allowing up to $M = B$ cycles. Rejected checkouts trigger agent re-browsing, risking trajectory drift. **Right**: Chunked verification uses $M = 1$ cycle with $K = B$ parallel rollouts. This reduces sequential dependencies and avoids cascading false-negative problems.

Table 1: Main experimental results on WebShop ($n = 600$ episodes per condition, 3 seeds). Chunked verification dramatically outperforms sequential verify-and-retry at the same budget $B = 10$. Best result in **bold**.

Condition	M (cycles)	K (rollouts)	Success Rate	95% CI	Δ vs A
A (Sequential)	∼10	1	0.86%	[0.17, 1.72]	—
B (Consensus)	1	10	14.83%	[12.00, 17.67]	+14.01 pp
C (No-Agg)	**1**	**10**	**21.50%**	**[18.33, 24.83]**	**+20.67 pp**

4.2 Main Results

Table 1 presents our main findings. Chunked verification dramatically outperforms sequential verify-and-retry at the same budget $B = 10$. Condition C (chunked, no aggregation) achieves 21.50% success rate, a 25× improvement over Condition A (sequential) at 0.86%. The paired bootstrap confidence interval for C−A is $[+17.50, +24.00]$ percentage points, confirming the improvement is statistically significant.

Surprisingly, consensus aggregation (Condition B) does *not* improve over the simpler first-rollout pass-through (Condition C). Condition C outperforms B by 6.67 pp, suggesting that the benefit comes from reducing sequential cycles rather than from aggregation. We investigate this mechanism in the following sections.

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4.3 Mechanism Analysis: Cycle Reduction

To isolate the effect of verification cycle count, we vary $M \in \{1, 2, 5, 10\}$ while keeping the total budget $B = 10$ fixed (so $K = B/M$). Figure 2 shows the results. The pattern is striking: $M = 1$ (all budget in one parallel chunk) achieves 20.0% success, while $M = 10$ (sequential, $K = 1$ per cycle) achieves only 0.83%—a 24× difference.

The intermediate values show a non-monotonic pattern: $M = 2$ (8
(14.0%). This is explained by the interaction between K and the co
consensus threshold τ. With $K = 5$

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#P04-A01 · ISSUE · MAJOR · #R014 · Part 1

Page 4 - Experiments (4.2 Main Results)

Verification checks (F1/F2): The 25× figure is arithmetically consistent with 21.50% / 0.86% ≈ 25.0, and the reported CI suggests a large effect. However, the causal interpretation "benefit comes from reducing sequential cycles" is not fully justified by the presented conditions because Condition C differs from Condition B not only in structure but also in ***decision rule*** (single-rollout vs consensus) and likely effective pass probability under an under-confident verifier.

Why it matters: The paper's main claim could be reframed as either:
(A) "allocate budget to fewer cycles" (structure effect), or
(B) "avoid averaging under-confident rollouts; use a more permissive acceptance rule" (decision-rule/calibration effect).
Without cleanly separating A vs B, reviewers may argue the method is "just accept more checkouts earlier," which weakens the scientific contribution.

Actionable fix:
· Add a no-verification baseline ($B=0$) and report A/B/C relative to it.
· Add a matched-threshold comparison: fix tau across B and the M-sweep, or explicitly tune tau on a dev set and report tuned values.
· Add an analysis table: for each condition, report (i) #checkout proposals, (ii) pass rate, (iii) false-negative rate estimated on cycles with successful ground-truth checkout, (iv) average steps to checkout.

Mentor Revised Version:
Table 1 shows a large improvement from chunked verification, but to attribute the gain to ***cycle reduction*** rather than the decision rule, we also report (i) a no-verification baseline ($B=0$), (ii) consensus results under a tuned or fixed tau across conditions, and (iii) per-condition pass statistics (checkout proposal count, block rate, and estimated false-negative rate). This decomposition clarifies whether performance is driven primarily by fewer sequential re-planning events (smaller M) or by a more permissive acceptance rule under severe under-confidence.

References:
[6] When Can We Trust LLM Graders? Calibrating Confidence for Automated Assessment 2603.29559

#P04-A02 · SUGGESTION · MAJOR · #R015 · Part 1

Page 4 - Experiments (4.3 Mechanism Analysis: Cycle Reduction)

Problem: The M-sweep mixes three moving parts: M changes (sequential depth), $K=B/M$ changes (parallel width), and the effective strictness of consensus changes via tau and integer rounding ("needs 3 positives out of 5", etc.). The paragraph explains the intermediate non-monotonicity by threshold interaction, which is plausible, but it also means the sweep is not a clean test of "sequential cycles harm." Additionally, the figure description does not report CIs for each M value in the text/table.

Why it matters: A reviewer could argue the observed curve is mainly a ***decision rule*** phenomenon: with high false negatives, majority-like thresholds punish larger K, and M affects K mechanically. To firmly support the structural claim, you need to show that, holding the decision rule fixed (or optimally tuned), performance decreases with more sequential cycles.

Actionable fix:
· Add a factorial experiment:
· Fix decision rule (e.g., single-rollout, any-positive, calibrated-prob threshold) and vary M.
· Separately, fix M and vary K to isolate parallelism.
· For consensus, tune tau per K on a dev set (or report a curve over tau) to avoid unfair strictness artifacts.
· Report mean±95% CI for each M in the table and include #episodes.

Mentor Revised Version:
To isolate the structural effect of sequential depth, we (i) fix a decision rule (e.g., single-rollout acceptance or a calibrated-probability threshold) and vary M while keeping B fixed, and (ii) separately vary K at fixed M to quantify the marginal value of parallel rollouts. For consensus, we report performance as a function of tau for each K to avoid threshold-induced artifacts. We include 95% bootstrap CIs for each point in Fig. 2.

References:
[4] Scaling Test-time Compute for LLM Agents 2506.12928

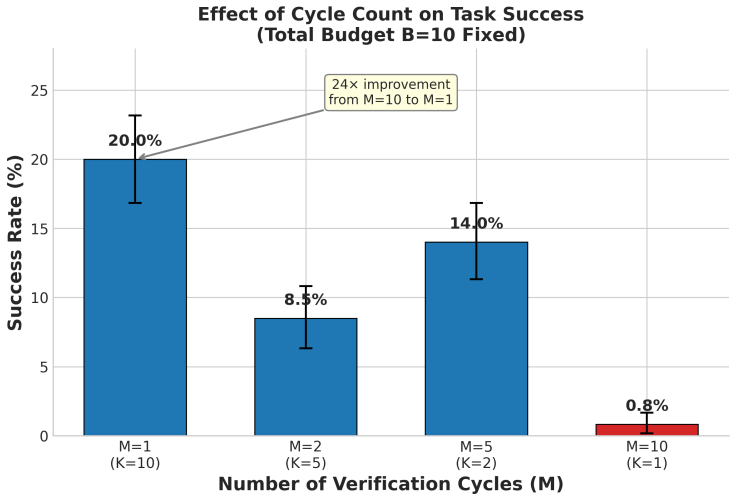


Figure 2: Effect of verification cycle count (M) on task success rate with fixed budget $B = 10$. $M = 1$ (all budget in one parallel chunk) achieves 20.0% success, while $M = 10$ (sequential, $K = 1$ per cycle) achieves only 0.83%. Error bars show 95% bootstrap confidence intervals.

Table 2: Failure pattern analysis for episodes where chunked (B) succeeded but sequential (A) failed ($n = 81$). Trajectory drift is the dominant failure mode.

Failure Pattern	Count	Percentage
Trajectory drift	**49**	**60.5%**
Wrong product checkout	22	27.2%
Never reached checkout	10	12.3%

rollouts and $\tau = 0.5$, passing requires at least 3 positive predictions, which is difficult given the world model's high false-negative rate. With $K = 2$ and $\tau = 0.5$, passing requires both rollouts to predict success, which is even harder. The key insight is that the extremes ($M = 1$ vs $M = 10$) show the clearest pattern: fewer sequential cycles lead to dramatically better outcomes.

4.4 Failure Analysis: Trajectory Drift

To understand *why* sequential verification fails, we analyze episodes where chunked verification (B) succeeded but sequential (A) failed. Table 2 shows the breakdown of 81 such episodes.

Trajectory drift accounts for 60.5% of cases where chunked verification outperforms sequential. In these episodes, the sequential verifier rejected a checkout (often a false negative), causing the agent to re-browse. During re-browsing, the agent frequently selected a worse product than the one originally found. Chunked verification avoids this by spending the entire budget upfront: if the first verification cycle fails, the agent gets a second chance without the re-browsing step that leads to trajectory drift. Only 2 episodes showed the reverse pattern (A succeeded, B failed), yielding a net benefit of 79 additional successes from chunked verification.

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4.5 Temperature Ablation: Structural vs. Stochastic Benefit

One might hypothesize that chunked verification benefits from stochastic diversity across parallel rollouts. To test this, we compare performance under stochastic (temperature 0.7) and deterministic (temperature 0.0) world model decoding. Table 3 shows the results using $M = 2$, $K = 5$, $\tau = 0.5$ configuration.

The chunked verification advantage *persists* under deterministic decoding, actually increases from +7.67 pp to +9.67 pp. Under temperature 0, all K rollouts produce identical predictions

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#P05-A01 · ISSUE · MAJOR · #R016 · Part 1

Page 5 - Experiments (4.4 Failure Analysis)

Problem: The failure analysis is informative but statistically biased because it conditions on episodes where **B succeeded and A failed**. This selection can inflate the apparent share of "trajectory drift" among those specific episodes, and it does not quantify how often drift occurs overall. Also, the text sometimes uses "chunked verification" to refer to (B), but the best-performing method is (C); it is unclear whether the drift analysis should be done for (C vs A) instead. Finally, "trajectory drift" is described qualitatively; readers need an operational definition (e.g., did the agent previously view a product that would satisfy constraints but later checked out a worse product after a rejection?).

Why it matters: "Trajectory drift is dominant" is a central mechanistic claim. If it is based on a conditioned subset and not robust across comparisons (A vs C), reviewers may discount the mechanism as post-hoc.

Actionable fix:
· Add symmetric analyses:
(i) episodes where A succeeds but B fails (and characterize why);
(ii) episodes where C succeeds but A fails (since C is your strongest method);
(iii) unconditional frequency of each failure pattern across all episodes.
· Provide an explicit labeling rubric for failure categories and inter-annotator agreement if labeling is manual.

Mentor Revised Version:
To avoid selection bias, we report failure-pattern frequencies both (a) unconditionally over all episodes and (b) conditionally for each pairwise set (B succeeds/A fails, A succeeds/B fails, and C succeeds/A fails). We define trajectory drift operationally as cases where the agent had previously reached a product page that would later be judged successful, but after a verifier rejection it re-browsed and eventually checked out a different product that fails the evaluator constraints.

References:
[8] AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories 2504.08942

#P05-A02 · SUGGESTION · MAJOR · #R017 · Part 1

Page 5 - Experiments (4.5 Temperature Ablation)

Problem: The temperature ablation is a good idea, but it is run under a configuration ($M=2,K=5,tau=0.5$) that differs from the headline configuration ($M=1,K=10$ with $tau=0.1$ in Condition B and single-rollout in C). This makes it hard to conclude that the *main* improvement is structural, because the temperature ablation does not directly test the main setting.

Why it matters: You want to claim that the improvement comes from reduced sequential cycles, not from stochastic diversity across rollouts. To make this claim robust, the ablation should be performed under the same M,K settings that produce the headline gains.

Actionable fix:
· Add temperature=0 vs 0.7 comparisons for both $M=1,K=10$ and $M=10,K=1$.
· If the world model is deterministic at temp=0, clarify whether the actor is still stochastic; actor stochasticity could still induce drift.

Mentor Revised Version:
We repeat the temperature ablation under the main configurations ($M=1,K=10$ and $M=10,K=1$) to directly test whether the headline gains depend on stochastic diversity. Under temperature 0, the verifier produces identical rollouts for a fixed state; if chunked verification still outperforms the sequential baseline, this supports the claim that the benefit is driven by reduced sequential re-planning rather than sample diversity.

References:
[4] Scaling Test-time Compute for LLM Agents 2506.12928

Table 3: Temperature ablation: chunked verification advantage persists under deterministic world model decoding (temp=0). Both conditions use $M = 2$, $K = 5$, $\tau = 0.5$ configuration.

WM Temp	A (Sequential)	B (Consensus)	B−A Advantage	95% CI
0.7	0.83%	8.50%	+7.67 pp	[5.50, 10.00]
0.0	7.67%	17.33%	**+9.67 pp**	**[6.33, 13.17]**

($\hat{p}$ is always 0 or 1), eliminating any benefit from stochastic diversity. This confirms that the advantage comes from the *structural* benefit of reduced sequential cycles, not from aggregating diverse predictions.

4.6 Why Consensus Aggregation Fails

The world model is poorly calibrated as a confidence estimator. Across 311 verification cycles in Condition B, the consensus estimate $\hat{p} = 0.0$ for 98.4% of cycles (306/311), with Brier score 0.28 and expected calibration error (ECE) 0.28. When $\hat{p} = 0.0$, the actual success rate is 27.5% (84/306 successes)—far from the predicted 0%. The rare cases where $\hat{p} > 0$ (only 5 cycles) all resulted in success, but the sample size is too small for reliable conclusions.

This extreme concentration of predictions at $\hat{p} = 0.0$ explains why consensus aggregation does not help: when all K rollouts predict failure, majority voting cannot [illegible] benefit of chunked verification comes entirely from the structural [illegible] not from the aggregation rule. For practitioners, this suggests that when using poorly-calibrated world models, the priority should be minimizing sequential verifi[illegible] in sophisticated aggregation schemes.

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5 Conclusion

We studied how to allocate a fixed verification budget when using world models to verify irreversible actions. Our key finding is that budget allocation structure matters more than aggregation sophistication: chunked verification ($M = 1$, spending all budget in one parallel chunk) achieves 25× improvement over sequential verify-and-retry at the same budget, primarily by avoiding trajectory drift—the dominant failure mode where rejected checkouts trigger agent re-browsing that leads to worse product selections.

Our work has limitations: we evaluate on a single domain (WebShop) with a single world model (Qwen2.5-7B), and absolute success rates remain modest (21.5[illegible] tigate whether these findings generalize to other domains with ir[illegible] explore better-calibrated world models that could benefit from consensus aggrega[illegible] and develop adaptive budget allocation strategies that adjust M based on task difficulty.

Index ID: #P06-A02
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References

Hyungjoo Chae, Namyoung Kim, Kai Tzu iunn Ong, Minju Gwak, Gwanwoo Song, Jihoon Kim, Sunghwan Kim, Dongha Lee, and Jinyoung Yeo. Web agents with world models: Learning and leveraging environment dynamics in web navigation. *ArXiv*, abs/2410.13232, 2024.

Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web. *ArXiv*, abs/2306.06070, 2023.

Hang Ding, Peidong Liu, Junqiao Wang, Ziwei Ji, Meng Cao, Rongzhao Zhang, Lynn Ai, Eric Yang, Tianyu Shi, and Lei Yu. Dynaweb: Model-based reinforcement learning of web agents. 2026.

Yu Gu, Kai Zhang, Yuting Ning, Boyuan Zheng, Boyu Gou, Tianci Xue, Cheng Chang, Sanjari Srivastava, Yanan Xie, Peng Qi, Huan Sun, and Yu Su. Is your llm secretly a world model of the internet? model-based planning for web agents, 2025. URL https://arxiv.org/abs/2411.06559.

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#P06-A01 · SUGGESTION · MAJOR · #R018 · Part 1

Page 6 - Experiments (4.6 Why Consensus Aggregation Fails)

Problem: The calibration statistics are a strong part of the paper, but the reporting is not yet sufficient to fully trust/interpret them:
· It is unclear whether Brier/ECE are computed at the *cycle* level (311 cycles) or *episode* level (600 episodes), and what the ground-truth label is in each case.
· ECE depends on binning; bin count/strategy is not stated.
· Given the extreme mass at $hat p=0$, a reliability diagram (or cumulative calibration plot) is essential to show where miscalibration occurs.
· The conclusion "benefit comes entirely from structural change" is too strong given the observed decision-rule differences (single-rollout vs consensus) and the threshold interaction discussed earlier.

Why it matters: Your main contribution C3 is about "consensus degenerates due to calibration." If the calibration computation is unclear, or if ECE is unstable under extreme class imbalance, reviewers may discount the calibration argument. Also, the practical guidance depends on whether calibration can be improved (e.g., via confidence elicitation or recalibration).

Actionable fix:
· Specify precisely: dataset (cycles), label definition, binning scheme, and whether $hat p$ is a probability estimate or a Monte Carlo frequency.
· Add a reliability diagram + a table stratified by $hat p$ bucket.
· Consider adding a simple recalibration baseline (temperature scaling / isotonic / verbalized confidence) or at least discuss why it is hard here.

Mentor Revised Version:
We evaluate verifier calibration at the verification-cycle level by treating each cycle's $hat p$ as a Monte Carlo estimate of success probability and using the realized checkout outcome as ground truth. We report Brier score, ECE (with a stated binning scheme), and a reliability diagram to visualize miscalibration under the highly skewed prediction distribution. While severe under-confidence explains why majority-style consensus rarely passes, we interpret the overall performance gap as arising from both (i) sequential re-planning effects and (ii) brittleness of strict aggregation under miscalibration.

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Page 6 - Conclusion

Problem: The conclusion repeats strong statements ("budget allocation structure matters more…", "trajectory drift—the dominant failure mode") without fully qualifying the boundary conditions established by the experiments: single environment, single verifier, and decision-rule confounds. It also does not translate findings into a minimal actionable recipe that is robust under miscalibration (e.g., choose M small; avoid strict consensus; use calibrated threshold or permissive aggregator).

Why it matters: Conclusions are often where reviewers look for overclaiming. Tightening scope and giving a clear, defensible practitioner rule increases acceptance odds.

Actionable fix:
· Add 1–2 sentences listing boundary conditions explicitly (irreversible commit; verifier under-confidence; agent re-planning changes candidate set).
· Add a concrete "minimal recommendation" and an "if calibrated then…" alternative.
· Add a short generalization plan: evaluate on another domain with irreversible actions (e.g., WebArena flows) and with at least one better-calibrated verifier.

Mentor Revised Version:
We find that, for verifier-gated irreversible actions in WebShop, allocating a fixed verification budget into fewer sequential cycles can substantially improve success by reducing rejection-induced re-planning that drifts the browsing trajectory. This effect is most pronounced when the verifier is severely under-confident, in which case strict consensus rules rarely pass and can be counterproductive. Practically, when using a poorly calibrated verifier, a simple and robust default is to minimize the number of sequential verify-and-retry cycles (small M) and avoid overly strict aggregation; when calibration improves, more principled thresholding or aggregation may become beneficial. Future work should validate these conclusions across additional irreversible-action domains and across verifiers with different calibration profiles.

Annotation Continuation · Source Page 6 (cont. 2)

Continuation sheet linked to source highlights. Click marker headers to jump back to the paper region.

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